

MITS Kochi | Muthoot Institute of Technology and Science (Autonomous)

Department of Electronics & Communication Engineering

An Energy-Efficient Hardware Accelerator for Convolutional Neural Network on Zynq SoC

MiniProject Presentation

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Team Number : 06

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Outline

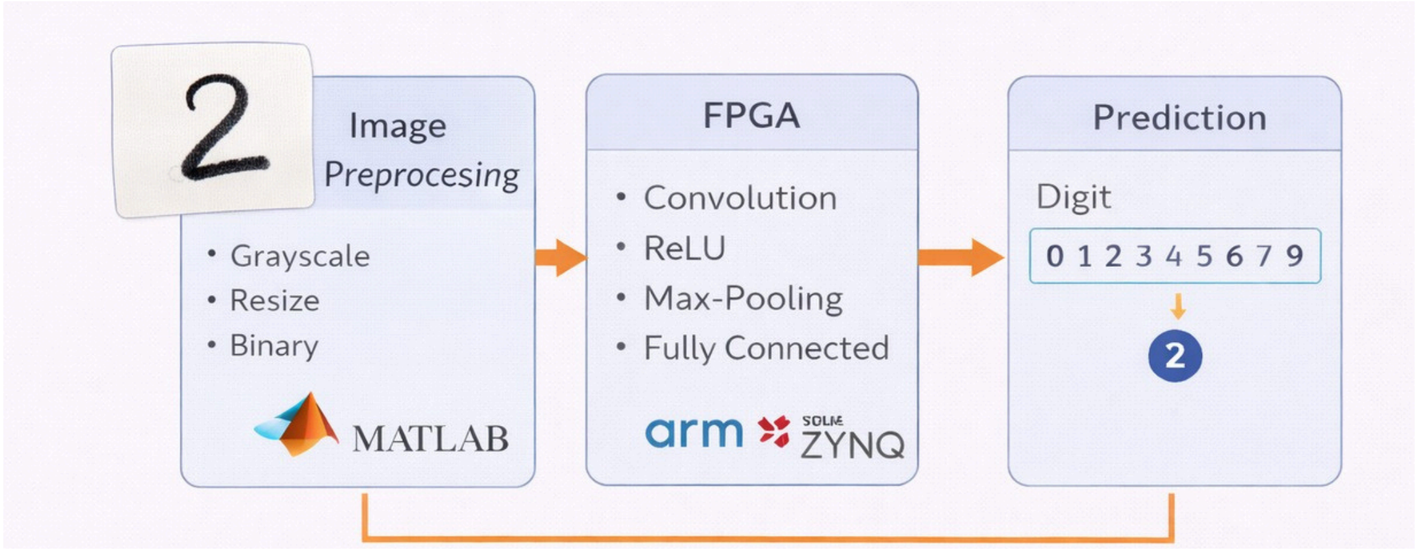
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Abstract

Artificial Intelligence applications are increasingly used in edge and embedded systems that require low-power hardware solutions. Convolutional Neural Networks (CNNs) provide accurate image recognition but involve high computation and memory usage. This project implements an energy-efficient CNN accelerator on an FPGA platform. Memory access is reduced using data reuse techniques, and computation is improved through parallel processing. ReLU activation and clock gating are used to reduce unnecessary operations and lower power consumption. The main objective of this project is to recognize and predict a single handwritten digit (0–9) in real time while maintaining low power and efficient hardware performance.

Introduction

- This project implements a CNN-based handwritten digit recognition system.
- The goal is to predict a single digit (0–9) in real time.
- MATLAB is used for image preprocessing (grayscale, resize, binary).
- The CNN computation runs on FPGA for faster and low-power processing.
- The network includes Convolution, ReLU, Max-Pooling, and Fully Connected layers.
- The design is implemented using Xilinx tools on a Zynq FPGA platform.



Motivation & Scope

- Primary goal: to recognize and predict a single handwritten digit (0–9) entered by the user.
- Increasing demand for AI processing on edge devices.
- High power consumption in CPU/GPU-based CNN inference.
- Memory bandwidth limitations in embedded systems.
- Need for real-time classification with low energy usage.

Scope

- FPGA-based CNN inference acceleration.
- Energy optimization using clock-gating techniques.
- Edge applications such as smart cameras and IoT devices.
- Hardware–software co-verification using MathWorks tools (MATLAB/Simulink).

Problem Statement

Problem Statement

- CNNs require heavy computation due to repeated convolution operations
- CPU/GPU implementations consume high power
- Not suitable for embedded or edge devices
- Memory bandwidth limits system efficiency
- Need real-time classification with low power and memory

Proposed Solution

- Design FPGA-based CNN accelerator
- Recognize handwritten digits (0–9)
- ReLU removes negative values → reduces computation
- Max-Pooling reduces data size → improves efficiency

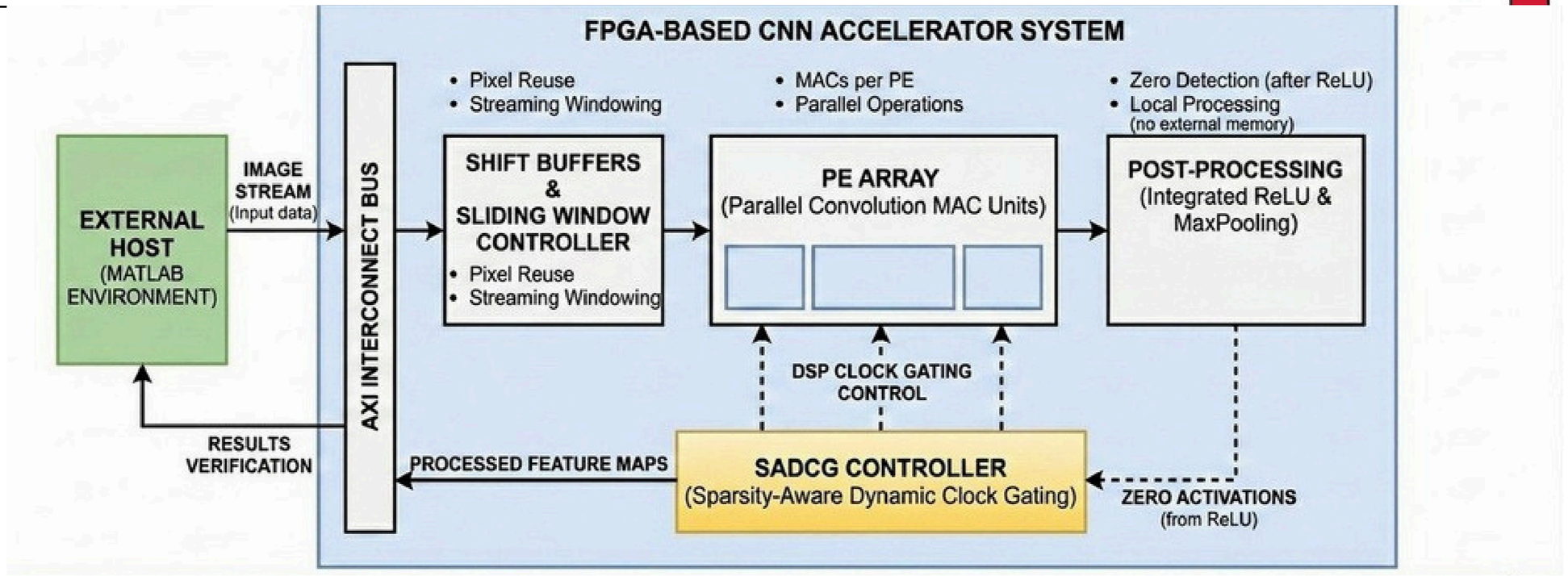
Gantt Chart

Task	Duration
Topic Selection	2 days
Study	5 days
Design	4 days
Model Setup	4 days
Hardware Design	5 days
Activation & Pooling	3 days
Power Saving	3 days
Integration	3 days
Testing	5 days
FPGA Work	6 days
Final Output	2 days

Methodology

- FPGA-based CNN accelerator using AXI interface
- Parallel convolution using PE array (MAC units)
- Shift buffers reduce memory access
- ReLU and MaxPooling integrated
- Dynamic clock gating reduces power
- MATLAB used for testing
- Pipelined architecture improves processing speed
- On-chip memory usage reduces external memory dependency

Methodology



Methodology

Convolution Operation

$$Y = X * W + b$$

Input

3	1	0
2	4	1
5	2	1

Filter

1	0	-1
1	-1	0
0	1	-1

Output

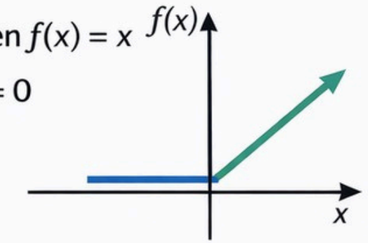
5	0	2
2	2	-1
1	2	

ReLU Activation

$f(x) = \max(0, x)$

ReLU: if $x > 0$, then $f(x) = x$

if $x < 0$, then $f(x) = 0$



Max-Pooling

2x2 Region

3	1	5
2	2	1

Max: 5

5

Argmax for Prediction

Scores: [2.1, 0.5, 6.8, 1.3, 0.7, 3.2, 0.1, 0.4, 1.1]

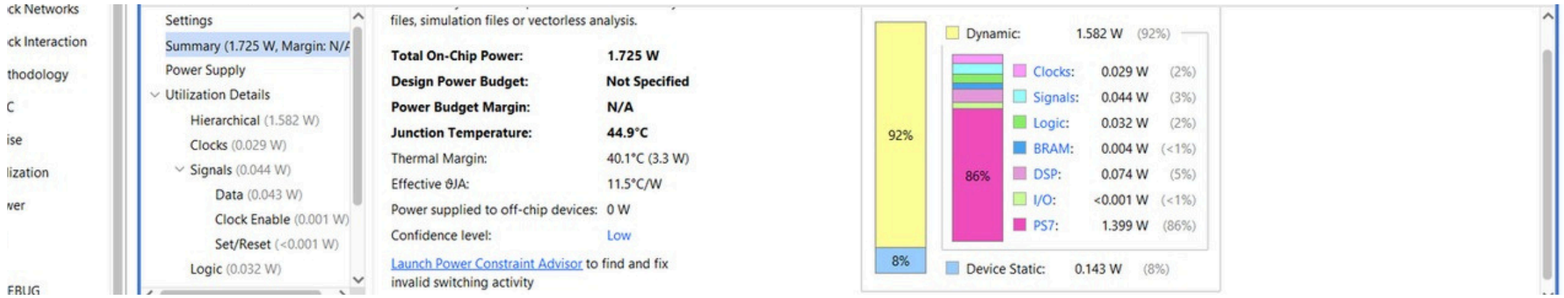
Prediction: **'2'**

Power consumption analysis

- Total Power Consumption: 1.725 W
- ARM processor consumes the majority of power
- FPGA hardware blocks consume comparatively less power

Reasons for Power Reduction

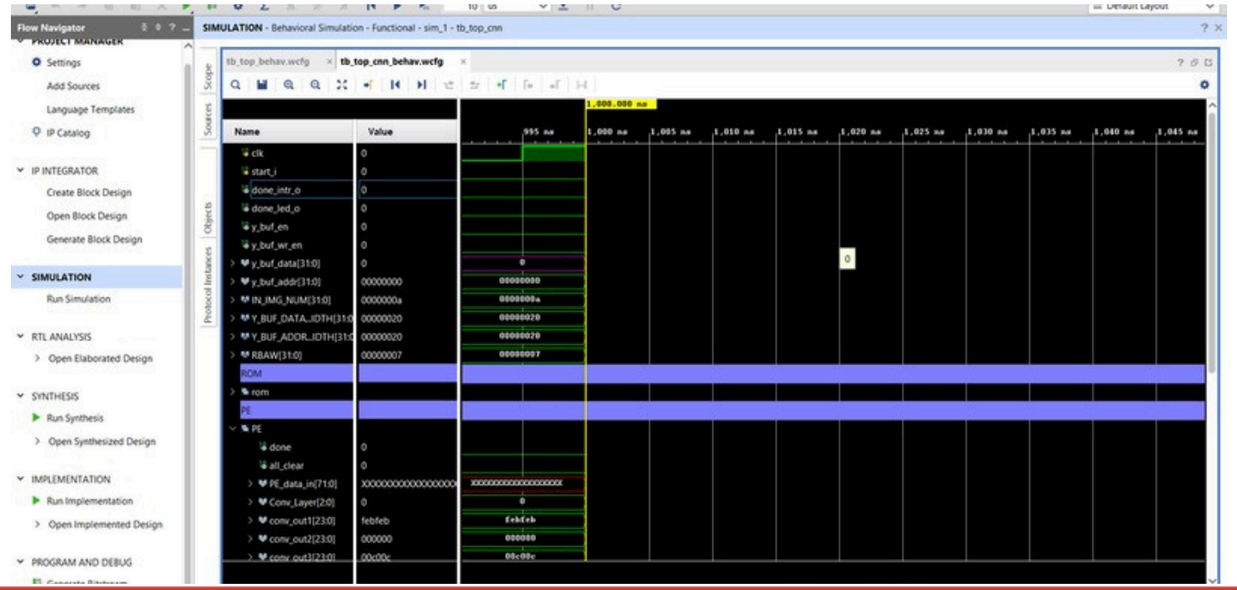
- ReLU converts negative values to zero → reduces computations
- Clock gating skips unnecessary operations
- Max-Pooling reduces data size and processing load

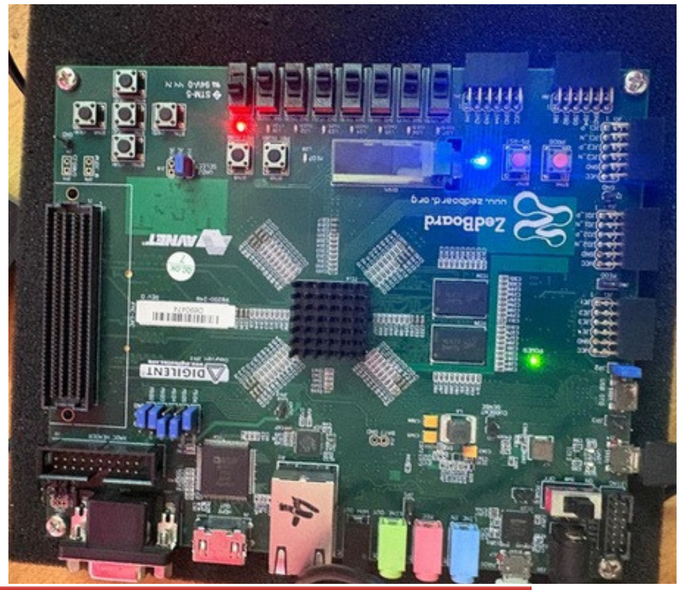


Results & Discussion

Observed Results

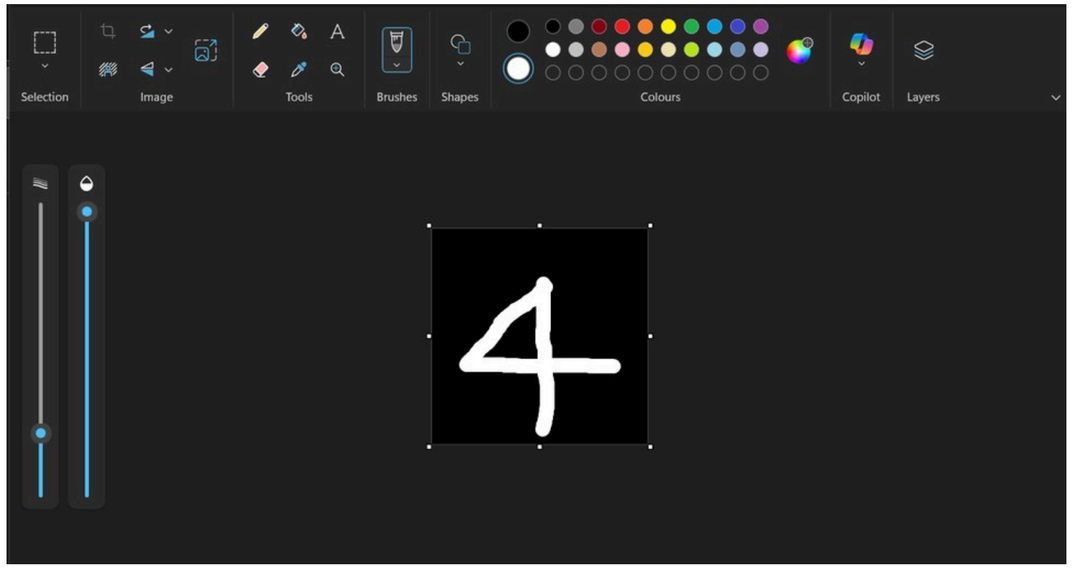
- CNN hardware successfully implemented on FPGA
- Convolution, ReLU, and control logic working correctly
- Proper data flow between processing elements and layers
- Intermediate CNN signals observed during execution
- Supports handwritten digit recognition (0–9)
- Suitable for real-time applications



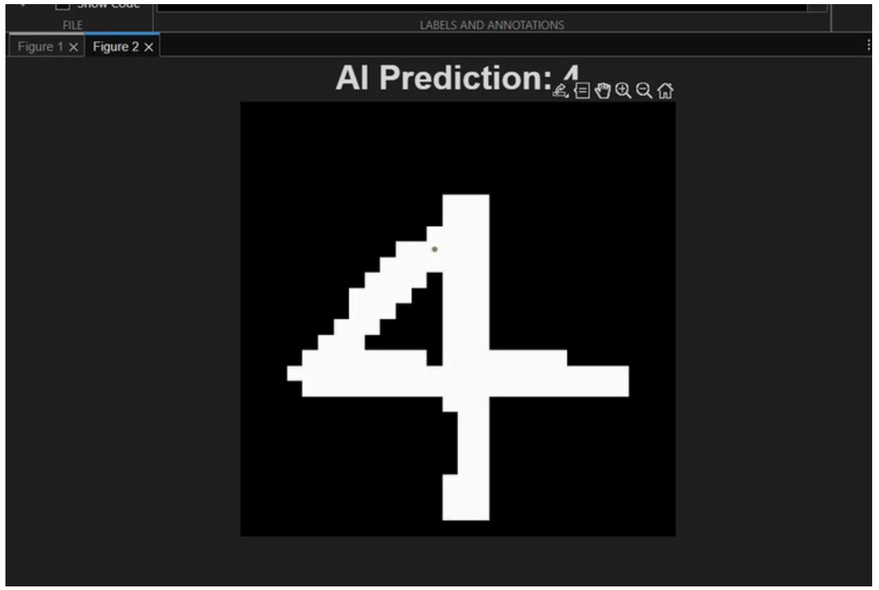


Results & Discussion

input given on MS paint:



Output:



Conclusion

This project implements a low-power CNN accelerator on an FPGA using a hardware–software co-design approach. The system successfully recognizes and predicts a single handwritten digit (0–9) in real time.
Techniques such as sliding-window buffering reduced memory access, and clock gating avoided unnecessary computations, lowering power consumption. The design showed efficient use of hardware resources and reliable operation, making it suitable for embedded and edge AI applications.

Future Scope

- Support more types of handwritten images
- Use deeper CNN networks for better performance
- Connect the system with IoT and smart monitoring systems
- Use in autonomous edge devices like robots and sensors
- Further reduce power consumption
- Take input directly from a camera instead of image files
- Improve prediction accuracy with better training
- Increase speed using more parallel processing
- Run the design on other or low-cost FPGA boards
- Add a simple user interface (GUI)
- Extend recognition from digits to letters/characters

References

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Sustainable Development Goals

Our Contribution:

SDG 7:

- Uses less power compared to normal systems
- Saves energy using efficient design techniques
- Suitable for low-power devices

SDG 9:

- Uses FPGA-based hardware (advanced technology)
- Improves speed and performance of AI systems
- Supports modern and smart applications